

# Miles V. Vollmer

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## EDUCATION & CERTIFICATIONS

B.S. in Computer Science with AI Concentration 2024 - Northeastern University, Boston, MA

**Certifications:** AWS Cloud Practitioner, BMC Certified Associate

## TECHNICAL SKILLS

**Cloud & Data Platforms:** Databricks (Lakeview/AI-BI, Unity Catalog, SQL Warehouses), AWS (EC2, S3), Grafana, Tableau, Power BI, dbt, BMC Helix, Dynatrace, Docker, Kubernetes, CI/CD, Git, NetSuite, Bash scripting

**Programming & Development:** Python, Java, SQL, R, JavaScript, TypeScript, HTML, CSS

**Automation & Performance Engineering:** ETL pipelines, BI platform migration, SQL transpilation, API reverse-engineering, agentic and multi-agent automation, system monitoring, release workflows, process optimization

**Machine Learning & AI:** PyTorch, TensorFlow, NLP, HuggingFace, Google ADK, multi-agent systems

**Domain Expertise:** Reinforcement learning, neural networks, data pipelines, agentic systems, LLM and RAG systems development, prompt engineering, full-stack development, web development and API usage, technical documentation for cross-functional audiences

## PROFESSIONAL EXPERIENCE

### Performance Engineering Consultant - Moviri

Jan 2025 – Present

Leading cloud migration, BI platform modernization, and enterprise efficiency engineering for major clients spanning banking (Lloyds Banking Group) and telecommunications (Fastweb).

**Lloyds Banking Group: BMC Helix Continuous Optimization (BHCO).** Led front-end and back-end development of a Business Units dashboard (multi-level drill-down from platform view to root cause via click navigation) that won the Innovation Award at the London BMC Helix Roadshow. Migrated client enterprise systems to the BMC Helix cloud platform and led capacity analysis to right-size ~10,000 overprovisioned servers, cutting \$2M+ per month in infrastructure spend. Extended a Dynatrace ETL collector to capture GCP host metadata (availability zone, project, instance type) across 2,300+ hosts using a multi-tier fallback resolution strategy, and built a Java HTTP client integrating BHCO's REST API with existing authentication infrastructure. Authored technical construction documentation for four reusable ETL extractor modules and their dashboard integrations.

**Fastweb: Grafana-to-Databricks BI Migration.** Architected and built a ~15-tool Python toolchain migrating a 451-dashboard, 4,150-panel Grafana estate (PostgreSQL/Redshift-backed) into Databricks AI/BI (Lakeview); covering complexity scoring, SQL transpilation, spec-driven dashboard assembly, and automated validation against source baselines; migrated over 400 dashboards to production to date. Reverse-engineered undocumented Lakeview platform internals (dynamic widget titles, conditional formatting, legend controls) with no public schema, turning each discovery into a reusable, tested build capability. Root-caused subtle data-correctness discrepancies by querying source warehouses directly, and translated Redshift/PostgreSQL SQL to Databricks SQL at scale, resolving timezone bugs and Lakehouse Federation incompatibilities. Designed process guardrails for AI-assisted engineering at scale, including a 52-test automated regression suite protecting the build pipeline.

Also designed and piloted HOAO, an internal multi-agent framework (Google ADK) where agents write, test, and self-validate their own tools; built an automated CI/CD release pipeline for monthly releases; and developed a full-stack HR automation system plus a NetSuite-integrated onboarding agent.

### Research Assistant - Evolutionary Genomics Lab, Northeastern University, Boston, MA

May 2021 - Sep 2021, Sep 2022 - Aug 2024

Developed multiple machine learning classifiers (Multilayer Perceptron, Support Vector Machine, Random Forest, Lasso Regression, and more) to predict coral disease states from 16S rDNA bacterial gene sequencing data, achieving over 97% classification accuracy. Conducted feature importance analysis utilizing SHAP values to identify the most significant bacterial taxa contributing to disease prediction. Co-authored the publication (below) in Environmental Microbiology detailing the pipeline and findings.

### Data Research Analyst - Iafate Lab, Mass General Hospital (MGH), Boston, MA

Jan 2022 - Aug 2022

Developed an automated pipeline for Genome-Wide Association (GWA) analysis of COVID-19 data, significantly improving data processing efficiency for large-scale genetic analysis. Analyzed demographic variations in positivity rates using advanced data visualization, leading to actionable insights for healthcare interventions and policy.

## PUBLICATIONS

Selwyn, JD, Despard, BA, Vollmer, MV, Trytten EC, Vollmer, SV (2024). "Machine Learning based identification of putative coral pathogens in endangered Caribbean staghorn coral." Environmental Microbiology 26 (9): e16700.

[enviromicro-journals.onlinelibrary.wiley.com/doi/epdf/10.1111/1462-2920.16700](https://onlinelibrary.wiley.com/doi/epdf/10.1111/1462-2920.16700)

## SELECTED PROJECTS

**Coral Population Genomics Pipeline:** Snakemake/ANGSD population-genomics pipeline across 290 whole genomes (2 species, 3 Caribbean regions) on AWS; 2M+ SNPs called, FST and SMC++ demographic analysis, and a candidate thermal-tolerance locus identified via LFMM; ~70% cost savings via spot-instance optimization. Feeds an in-prep publication.

**No-Limit Texas Hold'em Poker Agent:** CFR/MCCFR and a REBEL-style self-play neural agent for 6-player poker.

**4-Player Chess AI:** Minimax engine with alpha-beta pruning for 2v2 chess; 100% win rate vs. random AI, ~50% vs. humans.

**Extracting Traffic Data with NLP:** Pipeline for Boston 311 data to detect and classify traffic violations for urban planning use.

**Image Processing System:** Java application supporting advanced filters and format conversions.

## OTHER INTERESTS

Soccer (Co-Captain of Northeastern FC in BSSL), Surfing, Scuba Diving, Chess, Traveling, Hiking